

## Milestone 1: Airline Customer Holiday Booking

### Introduction

Booking a holiday flight is often one of the most exciting parts of planning a trip. For customers, choosing an airline is about more than just price and schedules. It also reflects their expectations for comfort, reliability, and an overall smooth travel experience. For airlines, these booking choices provide important clues about customer preferences and ways to improve their services in a competitive market.

This project implements a comprehensive machine learning pipeline to predict passenger satisfaction. We analyze the Airline Customer Holiday Booking dataset to identify patterns in customer behavior, understand the factors that influence their decisions, and develop models that can help predict future travel choices.

### Dataset:

**AirlineScrappedReview.csv:**

**For this file only , use the cleaned version provided to you on the CMS.**

The AirlineScrappedReview.csv file serves as a central record of passenger reviews, capturing both structured details and free-form feedback about airline experiences. Each entry includes the *Passenger\_Name*, the *Flying\_Date*, and the *Route* taken, which together provide the context of the trip. Customer opinions are reflected through the *Rating* column, the *Verified* flag confirming authenticity, and the open-text fields *Review\_title* and *Review\_content*, which express experiences in the passengers' own words. Additional attributes like *Traveller\_type* and *Class* describe the purpose of travel and the cabin flown, offering useful dimensions for segmenting customer groups (economy, business etc.). Together, these columns give us both the facts of the booking and each customer's review.

### **Customer\_comment.csv:**

The Customer\_comment.csv file provides a rich view of passenger feedback alongside detailed flight and service information. Each row combines operational data such as *flight\_number*, *origin\_station\_code*, *destination\_station\_code*, and *scheduled\_departure\_date* with performance indicators like *arrival\_delay\_group*, *departure\_delay\_group*, *seat\_factor\_band*, and *fleet\_usage*.

To capture the customer perspective, the file includes open-text responses in *verbatim\_text* and *ques\_verbatim\_text*, which are paired with derived fields such as **sentiments** and **transformed\_text** for deeper analysis. Additional attributes like *loyalty\_program\_level*, *fleet\_type\_description*, *entity*, and *response\_group* give further context about the customer and the airline's service.

### **Passanger\_booking\_data.csv:**

The Passanger\_booking\_data.csv file captures detailed information about customer booking behavior and travel choices. Each record includes core booking details such as *num\_passengers*, *sales\_channel*, *trip\_type*, and *booking\_origin*, which highlight how and where bookings are made.

Planning habits are reflected through *purchase\_lead* (days before departure the ticket was booked), *length\_of\_stay*, and the scheduled *flight\_hour* and *flight\_day*. Travel logistics are described by *route* and *flight\_duration*, while customer preferences are captured in fields like *wants\_extra\_baggage*, *wants\_preferred\_seat*, and *wants\_in\_flight\_meals*.

Finally, the *booking\_complete* column indicates whether the booking process was successfully finished. Together, these attributes provide a comprehensive view of passenger decision-making, enabling analysis of demand patterns, ancillary service preferences, and the factors that influence whether a booking is completed.

### **Survey data\_Inflight Satisfaction.csv**

The Survey\_data\_Inflight\_Satisfaction\_Score.csv file contains detailed survey responses that measure passenger satisfaction with their in-flight experience. Core identifiers like *flight\_number*, *origin\_station\_code*, *destination\_station\_code*, and *scheduled\_departure\_date* provide the travel context.

Passenger feedback is captured through *question\_text*, *score*, and *satisfaction\_type*, which together outline the nature of the survey, the rating given, and the type of satisfaction being measured. Additional insights come from attributes such as

*arrival\_delay\_minutes*, *cabin\_name*, *seat\_factor\_band*, and *loyalty\_program\_level*, which link operational and service factors to survey outcomes.

Dataset Source: [Airline Customer Holiday Booking Dataset](#)

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## Project Objectives

### 1. Data Cleaning:

Remove unnecessary columns and handle null values/duplicates, if any.

### 2. Data-Engineering Questions

To search for trends and insights, and understand customer sentiment towards the airline in this data sampling, add a new column to the provided cleaned version of the AirlineScrappedReview.csv by conducting Sentiment Analysis on review content.

Use the `SentimentIntensityAnalyzer()` method from the Vader library.

From the given dataset, you should analyze the given CSV files to answer and visualize the reasoning for the following questions:

- What are the top 10 most popular flight routes? What is the distribution of bookings across flight hours?
- By analyzing review patterns, which traveler type and class combination yields the highest and lowest ratings?

### 3. Predictive Modeling Task

Develop a statistical ML model or shallow Feed-Forward Neural Network (FFNN) to predict passenger satisfaction (satisfaction) using features like:

- Traveller-related features such as type and class.
- Features about the flight.
- The verification of the review and the calculated sentiment score.

The target variable is derived from the Rating column, where passengers are classified as either:

- Satisfied if Rating  $\geq 5$
- Dissatisfied if Rating  $< 5$

This is a binary classification problem. The developed models should be evaluated using accuracy, precision, recall, and F1-score

#### 4. Model Explainability

To gain deeper insights into the model's behaviour, apply explainable AI techniques (SHAP, LIME) to interpret predictions, ensuring transparency about the most influential features.

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## Project Deliverables

1. A refined Jupyter Notebook documenting all steps and providing a reproducible workflow; Adding justifications and explanations to the design and results.
2. A cleaned dataset with the Sentiment\_Analysis column added, and modified columns after the feature engineering step (if any).
3. An analytical text report answering and visualizing the data engineering questions. The report should also include the features used in the predictive model and why each feature was selected.
4. Predictive model (statistical ML or shallow FFNN) for predicting passenger satisfaction .
5. XAI outputs (SHAP plots and LIME explanations) illustrate the contribution of different features to predictions.
6. An inference function that takes raw input and returns the model prediction.

## Submission and Deadline

Please submit your GitHub repository, fulfilling the specified requirements, using the following form: <https://forms.gle/GyFaidvUY2BoKCby6> by **October 22<sup>nd</sup> at 11:59 pm**

Ensure your repository remains private until the deadline. After the deadline, you will be required to make it public or add the course account (csen903w25-sys) as a collaborator for milestone grading.